IsoDEC – A program for exploring polycrystal elasticity

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web address: http://www.ncnr.nist.gov/programs/crystallography/software/downloads.html

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1. Introduction

IsoDEC has been available since 2012 for the calculation of diffraction elastic constants (DEC), stress, and strain (Gnäupel-Herold, 2012). The current version provides several new unique features, and major additions in functionality. IsoDEC calculates bulk elastic constants for aggregates consisting of up to three phases with and without preferred orientation and for all crystal structures. Grain shape parameters are included in the calculations. The functionality for materials with preferred orientation requires EBSD data or a user supplied orientation distribution function (ODF), typically as a text file output from MTEX (Hielscher & Schaeben, 2008). IsoDEC incorporates all relevant DEC models such as the lower/upper bound models Reuss/Voigt (Reuss, 1929; Voigt, 1928; Möller & Martin, 1939) and Kröner/inverse Kröner (Bollenrath, F., Hauk, V., Müller, 1967; Gnäupel-Herold *et al.*, 2012) as well as the Geometric Average (Matthies *et al.*, 2001), Hill type averages (Hill, 1952), and the modified Voigt model (Murray & Noyan, 1999). Among several unique features, IsoDEC can use EBSD grain data directly and utilize the information on grain shape and neighboring grains for each orientation and (hkl) in order to calculate DEC using the Kröner type models. Furthermore, IsoDEC allows the fitting of single crystal elastic constants using diffraction elastic constants as input. Major functions of IsoDEC are demonstrated through the Demo modes in the tables "DEC/Stress/Strain" and "SC Fit".

2. Isotropic calculations

The start page is the "Material data input form" tab which allows the calculation of isotropic constants s_1 and $\frac{1}{2}s_2$. They are used in the equation

$$\varepsilon_{\varphi\psi} = \tfrac{1}{2} s_2(hkl)\begin{bmatrix}(\sigma_{11}\cos^2\varphi + \sigma_{22}\sin^2\varphi + \sigma_{12}\sin 2\varphi)\sin^2\psi + \\ (\sigma_{13}\cos\varphi + \sigma_{23}\sin\varphi)\sin 2\psi + \sigma_{33}\cos^2\psi\end{bmatrix} + s_1(hkl)(\sigma_{11}+\sigma_{22}+\sigma_{33}) \qquad (1)$$

Under “Compound” one can select a material from a database (pfcryst.dat) included in IsoDEC.zip. This loads the single crystal elastic constants, crystal symmetry, lattice parameters (important for non-cubic materials) and the (isotropic) bulk elastic constants E, nu, K, G, only two of which are independent. Then fill in values for H, K, L and click “Calc DEC”. The calculation is based on the Kröner model [3,7,10].

3. Entering new material data

If the elastic constants C_{ij} are not in the database then new values can be entered and stored in pfcryst.dat (pfcryst.dat is a textfile and can be edited). This is done using these steps:

a) Enter a name in the “Compound” field. Chose the crystal symmetry from the drop box.
b) Enter the single crystal elastic constants
c) On the top left corner, click Calculate Macro - Constants.
d) Enter the unit cell parameters. Particularly calculations for non-orthogonal unit cells rely on correct unit cell parameters!
e) Optional: Enter a literature reference.
f) Click Add to Database.

The entry field for “hkl of common reflections” is optional. Entries made here are used by the PF neutron data analysis software used at the NIST Center for Neutron Research.

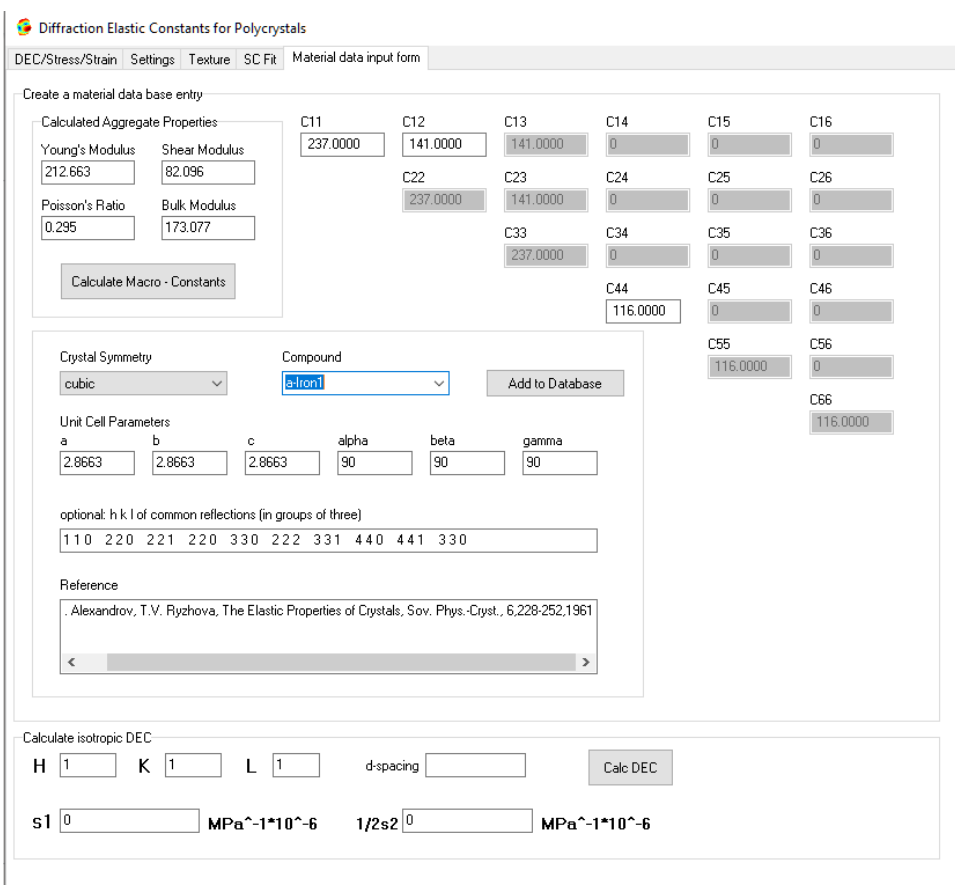
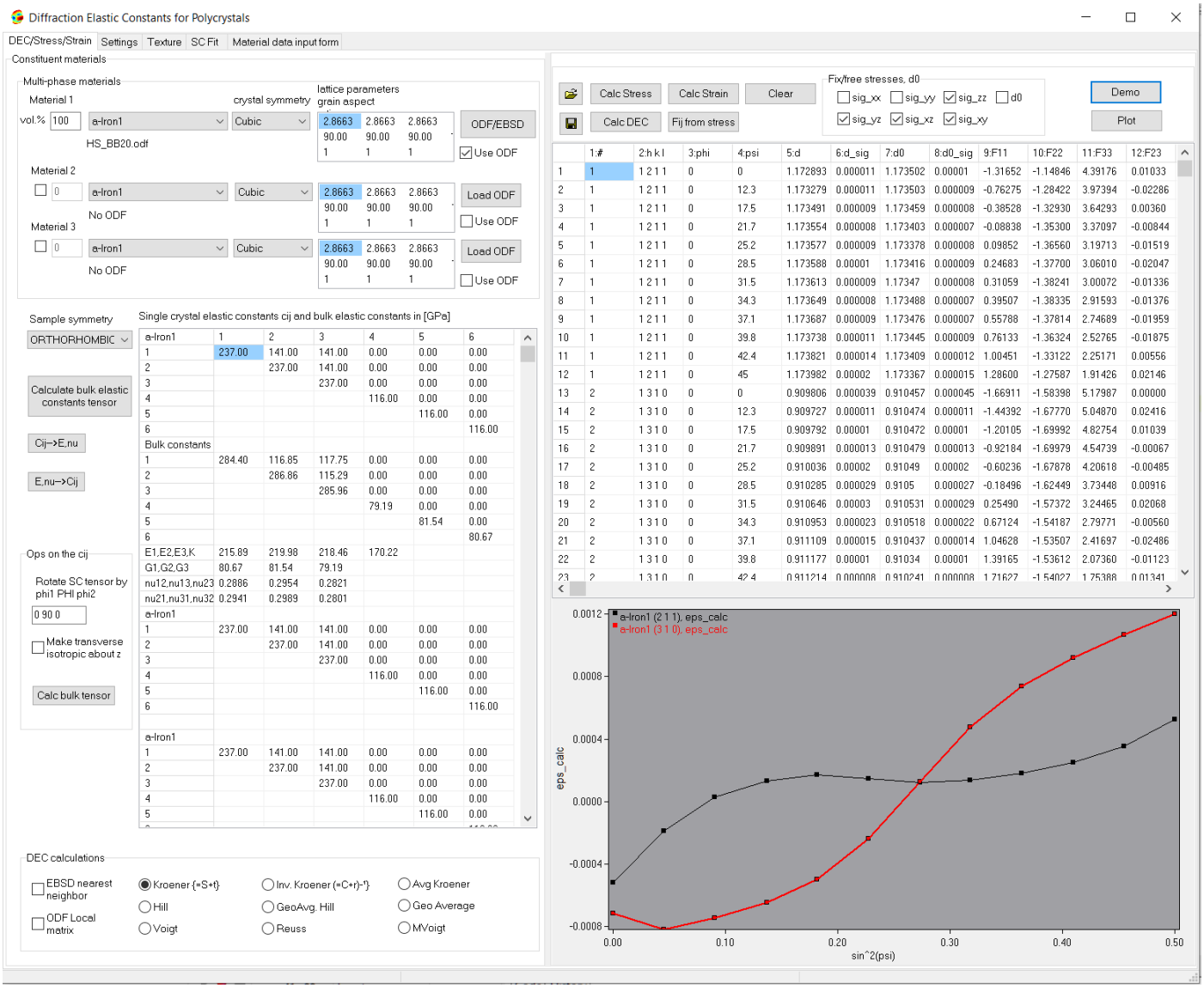


Figure 1: Screenshot of IsoDEC (Material data input form).

4. DEC/Stress/Strain

The calculation of bulk elastic constants (left), diffraction elastic constants, stress and strain (middle, top) is available in this tab.



4.1.Bulk elastic constants

Depending on which micro-mechanical model is selected, bulk constants are calculated in two different ways. Kröner type models use a set of 4572 random orientations for each phase in an iteration such that the bulk tensors for compliance and stiffness minimize the elastic polarization and elastic susceptibility (Kröner, 1958). All other models are based on numerical integration over grain orientations. Up to three constituent phases can be included, each with its own ODF, loaded as a 4-column (phi1,PHI,phi2, weight) text file. Euler angles are given in Bunge notation. Material data such as bulk and single crystal elastic constants and lattice parameters are stored in the data base and can be loaded from the drop-down lists. Whether the ODF is included in calculations is controlled by checking the "Use ODF" box. Volume fractions can be entered on enabling the material (check boxes on the left). Only material 1 can be used to read EBSD data. Each material can be assigned an average grain shape in the third row of the lattice parameter table:

lattice parameters
grain aspect ratios

2.8663	2.8663	2.8663
90.00	90.00	90.00
10.5	10.5	1.0

Only Kröner type models are sensitive to the grain shape through the Eshelby-tensor; however, the grain shape effect on bulk elastic constants is generally weak. Exceptions are composite materials that contain high percentages of oriented fiber bundles.

Aside from the general importance of the bulk moduli they also affect DEC models that contain a grain-matrix interaction (Kröner type models and the geometric average). The larger the difference between single crystal moduli and bulk moduli the larger the effect on the DEC. One prominent example are aluminum metal matrix composites in which small fractions of elastically hard particles such as silicon carbide are surrounded by an elastically much softer matrix of aluminum alloy, having an appreciable effect of the effective diffraction elastic constants of silicon carbide in the composite.

Single crystal elastic constants cij and bulk elastic constants in [GPa]

a-Iron1	1	2	3	4	5	6
1	237.00	141.00	141.00	0.00	0.00	0.00
2		237.00	141.00	0.00	0.00	0.00
3			237.00	0.00	0.00	0.00
4				116.00	0.00	0.00
5					116.00	0.00
6						116.00
Bulk constants						
1	282.39	118.16	118.16	0.00	0.00	0.00
2		282.39	118.16	0.00	0.00	0.00
3			282.39	0.00	0.00	0.00
4				82.11	0.00	0.00
5					82.11	0.00
6						82.11
E1,E2,E3,K	212.68	212.68	212.68	172.90		
G1,G2,G3	82.11	82.11	82.11			
nu12,nu13,nu23	0.2950	0.2950	0.2950			
nu21,nu31,nu32	0.2950	0.2950	0.2950			
a-Iron1						
1	237.00	141.00	141.00	0.00	0.00	0.00
2		237.00	141.00	0.00	0.00	0.00
3			237.00	0.00	0.00	0.00
4				116.00	0.00	0.00
5					116.00	0.00
6						116.00
a-Iron1						
1	237.00	141.00	141.00	0.00	0.00	0.00
2		237.00	141.00	0.00	0.00	0.00
3			237.00	0.00	0.00	0.00
4				116.00	0.00	0.00
5					116.00	0.00
6						116.00

Calculate bulk elastic constants tensor

This table lists the single crystal constants (in GPa) for each phase, with material 1 at the top, followed by the bulk constants and values for Young's modulus E, Poisson's ratio nu, the bulk modulus K and the shear modulus G, with indices 1,2,3 indicating the principal direction for anisotropic bulk constants. Any two or three phase composite can be constructed by selecting the material, their (optional) ODF and grain shape (default is spherical), followed by clicking. The user should also select what sample symmetry is expected. Lower sample symmetry will cause some calculations to slow down.

Cij-->E,nu

E,nu-->Cij

These two buttons calculate bulk moduli from the tensor representation of the bulk constants and vice versa. Anisotropic bulk constants yield three moduli for the three principal axes and six values for Poisson's ratio (two values for each of the two directions perpendicular to the applied stress)

Ops on the cij

Rotate SC tensor by phi1 PHI phi2

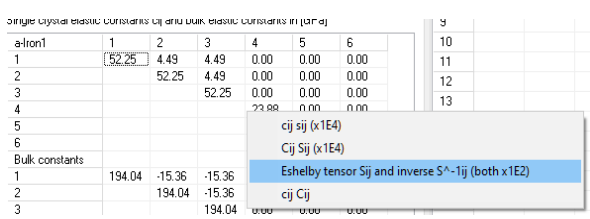
0 90 0

☐ Make transverse isotropic about z

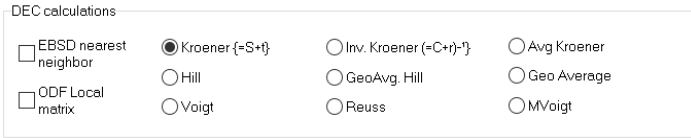
Calc bulk tensor

This field allows to perform a rotation of the single crystal elastic constants of material 1, and write the results as bulk constants. Additionally, checking the box 'Make transverse isotropic' causes the rotated tensor to be averaged about the z-axis. These options allow the explorations of upper and lower limits of elastic properties of highly textured materials.

Right-clicking on the tables and the plot window open other menu selections:



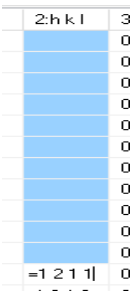
The four options in this menu switch between stiffnesses (in GPa) and compliances (in 10^{13} Pa^{-1}) as well as the Eshelby tensor S and its inverse S^{-1}. This applies only to Material 1. Lower case "c" or "s' indicate single crystal tensors while uppercase "C" or "S" denote bulk tensors.



The box on the bottom left controls which grain interaction model is used. With the exception of "EBSD nearest neighbor" and "ODF local matrix" the model selection also determines how the bulk elastic constants are calculated from the single crystal constants. For example, the classic upper/lower bounds on the bulk elastic constants are obtained if either the Voigt or Reuss model is selected.

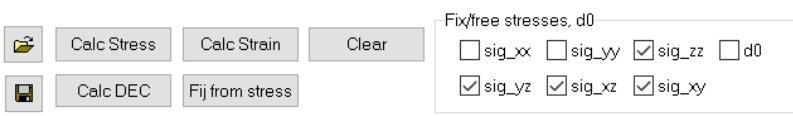
4.2.DEC/Stress/Strain

4.2.1. Editing



Each of the three modes Stress/Strain/DEC needs its input provided in this table. Input data can be loaded from previously saved data, pasted from a spreadsheet such as EXCEL, or entered manually. Repeating values can be entered by selecting a range of cells and typing "=", followed by the characters or numbers to be used. Range fill is activated by pressing Crtl+Enter.

4.2.2. Stress



In the "Fix/free stresses" box above, checking a box will fix the respective stress component to zero in the fit. Checking the "d0" box will force the use of d0-values given in column 7.

In order to calculate stresses each of the columns in the table below must be populated:

	1:#	2:h k l	3:phi	4:psi	5:d	6:d_sig	7:d0	8:d0_sig	9:F11	10:F22	11:F33	12:F23	13:F13	14:F12
1	1	1 2 1 1	0	0	1.172893	0.000011	1.173502	0.00001	-1.26227	-1.26227	4.45174	0.00000	0.00000	0.00000
2	1	1 2 1 1	0	12.3	1.173275	0.000011	1.173503	0.000009	-1.00295	-1.26227	4.19242	0.00000	2.37871	0.00000
3	1	1 2 1 1	0	17.5	1.173491	0.000009	1.173459	0.000008	-0.74558	-1.26227	3.93505	0.00000	3.27753	0.00000
4	1	1 2 1 1	0	21.7	1.173554	0.000008	1.173403	0.000007	-0.48109	-1.26227	3.67056	0.00000	3.92615	0.00000
5	1	1 2 1 1	0	25.2	1.173577	0.000009	1.173378	0.000009	-0.22638	-1.26227	3.41585	0.00000	4.40286	0.00000
6	1	1 2 1 1	0	28.5	1.173588	0.00001	1.173416	0.000009	0.03870	-1.26227	3.15076	0.00000	4.79232	0.00000
7	1	1 2 1 1	0	31.5	1.173613	0.000009	1.17347	0.000006	0.29768	-1.26227	2.89178	0.00000	5.09137	0.00000
8	1	1 2 1 1	0	34.3	1.173649	0.000008	1.173488	0.000007	0.55228	-1.26227	2.63718	0.00000	5.32022	0.00000
9	1	1 2 1 1	0	37.1	1.173687	0.000009	1.173476	0.000007	0.81683	-1.26227	2.37264	0.00000	5.49829	0.00000
10	1	1 2 1 1	0	39.8	1.173736	0.000011	1.173445	0.000009	1.07899	-1.26227	2.11047	0.00000	5.62030	0.00000
11	1	1 2 1 1	0	42.4	1.173821	0.000014	1.173409	0.000012	1.33580	-1.26227	1.85367	0.00000	5.69066	0.00000
12	1	1 2 1 1	0	45	1.173982	0.00002	1.173367	0.000015	1.59473	-1.26227	1.59473	0.00000	5.71418	0.00000
13	2	1 3 1 0	0	0	0.909806	0.000039	0.910457	0.000045	-1.66377	-1.66377	5.25475	0.00000	0.00000	0.00000
14	2	1 3 1 0	0	12.3	0.909727	0.000011	0.910474	0.000011	-1.34980	-1.66378	4.94078	0.00000	2.88014	0.00000
15	2	1 3 1 0	0	17.5	0.909792	0.00001	0.910472	0.00001	-1.03818	-1.66378	4.62915	0.00000	3.96843	0.00000
16	2	1 3 1 0	0	21.7	0.909891	0.000013	0.910479	0.000013	-0.71793	-1.66378	4.30891	0.00000	4.75378	0.00000
17	2	1 3 1 0	0	25.2	0.910036	0.00002	0.91049	0.00002	-0.40953	-1.66378	4.00051	0.00000	5.33099	0.00000

Column 1: Identifier. Each stress tensor needs its own identifier, e.g. a location on or in the sample. If one wishes to calculate stresses based on different sets of d-spacings then each set should get its own identifier. For the same identifier but different (hkl) d0 must be fixed.

Column 2: Four integers needed. The first number is the material number and it can be 1,2 or 3. The following three integers are the h k l.

Column 3: Azimuth angle phi as part of the orientation of the scattering vector in spherical polar coordinates.

Column 4: tilt angle psi as part of the orientation of the scattering vector. One has X=(phi=0,psi=90), Y=(phi=90,psi=90) and Z=(phi=0,psi=0).

Column 5: Measured d-spacing in Angstroms.

Column 6: Uncertainty of measured d-spacing. Only non-zero numbers are acceptable.

Column 7: Unstressed d-spacing, either from measurement or otherwise obtained.

Column 8: Uncertainty of d0. Only non-zero numbers are acceptable.

Column 9-14: Diffraction Elastic constants (DEC) specific for the (hkl) and orientation (phi,psi), calculated from a user-selected model. The meaning of the F_{ij} is straightforward. They represent the slopes of lattice strain over applied stress $\Delta\varepsilon(hkl,\varphi,\psi)/\Delta\sigma_{applied}$, in units of TPa^{-1} for a given reflection (hkl) at a sample orientation (φ,ψ). The subscripts "ij" indicate the direction of the applied stress. For example, F_{11}(211,0,45) is the slope for the (211) reflection, with stress applied in the "11" or x-direction, and the measurement direction (scattering vector) at a tilt angle of 45°. The use of DEC is the most direct way to evaluate stress from strain through the equation

$$\varepsilon_{\phi\psi} = \frac{d_{\phi\psi} - d_0}{d_0} = F_{11}(hkl,\varphi,\psi)\sigma_{11} + F_{22}(hkl,\varphi,\psi)\sigma_{22} + F_{33}(hkl,\varphi,\psi)\sigma_{33} +$$

$$2F_{12}(hkl,\varphi,\psi)\sigma_{12} + 2F_{13}(hkl,\varphi,\psi)\sigma_{13} + F_{23}(hkl,\varphi,\psi)\sigma_{23}$$

The set of n equations for n d-spacings can also be used to evaluate d0 (d0 box must be unchecked) if suitable boundary conditions can be applied such as σ_{33}=0, and if more measurements than unknowns are available. Calculated stresses and d0 are listed in columns 19-31.

Column 15: Contains calculated d-spacings from a stress fit.

Column 16: I(ODF) or ODF intensity in m.r.d.×100. This value is equal to the intensity of a normalized pole figure for (hkl) at the orientation (φ,ψ). For EBSD data

Column 17: Lattice strains obtained from the d-spacings in column 5 and the unstressed d-spacings in column 7.

Column 18: Identifier from column 1 to match the stresses/uncertainties in columns 19-30 to the d-spacings with the same identifier.

Column 19-30: Stresses and their uncertainties.

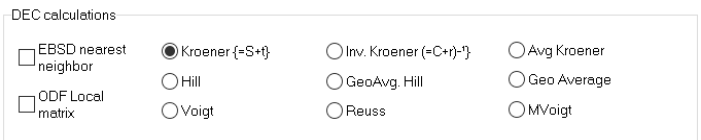
Column 31-32: Calculated unstressed d-spacings and their uncertainties. This column is populated only if d0 is freed in the stress fit. The calculated d0 is only meaningful if appropriate boundary conditions have been selected. For example, the data in the demonstration have only tilts ψ for φ=0 (X-Z plane). Therefore, leaving the stress σ_{yy} as a free parameter (box unchecked) will not give meaningful results because there are no data available in the Y-Z plane. Fixing σ_{zz}=0, however, allows the calculations of a d0 value such that σ_{zz}=0.

Column 33-36: Principal stresses σ_1, σ_2, σ_3 and the rotation angle gamma (only calculated for plane stress σ_{33}=0, σ_{i3}=0). Gamma is the angle by which the tensor σ_{ij} is rotated such that all components σ_{ij}=0 with i≠j. This output is useful if substantial magnitudes of shear stresses σ_{12} are present.

4.2.3. Diffraction Elastic Constants (DEC)

2:h k l	3:phi	4:psi
1 2 1 1	0	45

The calculation of diffraction elastic constants needs, at a minimum, column 2 to contain four numbers, separated by space (m h k l) as well as phi and psi in column 3 and 4."m" is the material number (1, 2, or 3) and "h k l" are the Miller indices. Optionally, three more parameters can be supplied in column 2 for the grain shape parameters (ellipsoid aspect ratios). The format is (m h k l a b c), all separated by space. DEC are calculated using the model selected from the box:



"**EBSD nearest neighbor**" will only work if EBSD data have been loaded. The EBSD data set is searched for grains oriented for diffraction for a given (hkl) and (phi,psi) within an acceptable misorientation window ('Settings' tab). The average grain shape for (hkl,phi,psi) is calculated; it is used for Kröner type models. The grains surrounding the diffracting grains within ≈1 grain diameter are identified, and the 'local' elastic properties of the neighboring grains are calculated using the geometric average. This 'local' matrix is used for DEC calculations with Kröner type models. Otherwise, each DEC model works as expected; however, the path integral average for rotations about the scattering vector /lattice plane normal is now replaced by a discrete average. The number of grains in this average depends on the acceptable misorientation angle set in 'Settings'. The number of grains found is calculated from sum of weights of diffracting grains divided by the average weight. The number is written in the column I(ODF) in the worksheet.

"**ODF local matrix**". An ODF must be loaded for this option. Checking this box will enable calculation of matrix constants (Kröner type models) using the average of grain orientations from the path integral. In essence, this assumes that diffracting grains are sounded by grains of similar orientation. This has the effect of reducing the grain-matrix interaction in Kröner type models.

4.2.4. Strain

The calculation of lattice strains needs the same input as stress. Additionally, stress values in column 19-29 must be supplied together with an identifier in column 18. The identifier distinguished between repeating orientations.

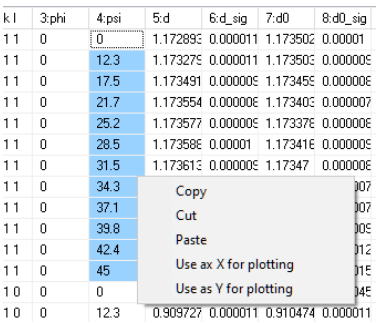
4.2.5. Fij from applied stress

Measurements of diffraction elastic constants are based on d-spacings measured at known levels of stress. Generally, for a given measurement direction (phi, psi) there should be at least two levels of applied stress to allow a linear regression of lattice strain vs. applied stress. The F_{ij} constant is the slope of that regression. This option needs d-spacings and their uncertainties (columns 5,6), unstressed d-spacings and uncertainties (columns 7,8), the measured direction phi and psi (columns 3,4), the applied stresses (column 19) and the identifiers in column 2 and column 18 in order to match the measurements at the same applied stress to the stress values in column 19. The measurement direction must have the correct relationship to the stress direction. For example, let's assume there are three applied stresses, each with 10 measurements in different directions (phi, psi). Column 18 has three rows containing identifiers 1, 2, 3. Column 19 has the three stresses, and column 2 has 10 rows with identifier 1, then row 11-20 with identifier 2, and row 21-30 with identifier 3. Measurements were done from the normal direction (phi=0,psi=0, perpendicular to the applied stress) in steps of 10 degree to (phi=0,psi=90, parallel to the applied stress). Output is written in column 9, containing two numbers as F11 and its uncertainty.

4.3. Demo mode

This will run through an example of two sets of d-spacings measured under known applied stresses using X-ray diffraction. These are data from actual measurements. An ODF obtained from neutron pole figure measurements will be loaded, followed by a DEC calculation. The following stress calculation will be increasingly refined through use of better DEC models and the use of grain shape parameters, leading to better agreement between calculated stresses and applied stresses.

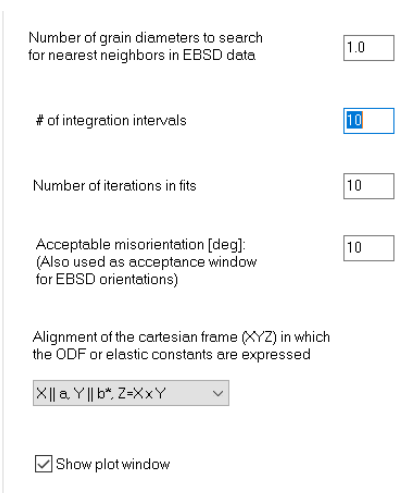
4.4.Plot



Plotting requires to select a cell range by holding the mouse button and dragging or Shift+arrow up/down. Right click will show a menu from which data can be designated as X or Y data. Once both X and Y data have been selected, clicking the Plot button will show data. Up to 10 X-Y plots can be displayed simultaneously. The plot window can be reset/cleared by right-clicking the plot area and selecting the option. If values in the designated cells change, simply click "Plot" again to show the change.

5. Settings

"**Number of grain diameters**": This applies to EBSD files and the search for grain orientations that, for the (hkl) and phi, psi specified in column 2, 3, 4 and in "DEC/Stress/Strain". A number greater than 1 increases the search radius for finding neighbor grains.



"**# of integration intervals**": Diffraction elastic constants and the anisotropic Eshelby tensor are calculated by numerical integration. In rare cases, lower symmetries or extreme grain shapes may cause small numerical errors ($\approx 10^{-2}$). This can be alleviated by increasing the number of integration intervals.

"**Number if iterations in fits**": This applies to fitting single crystal constants.

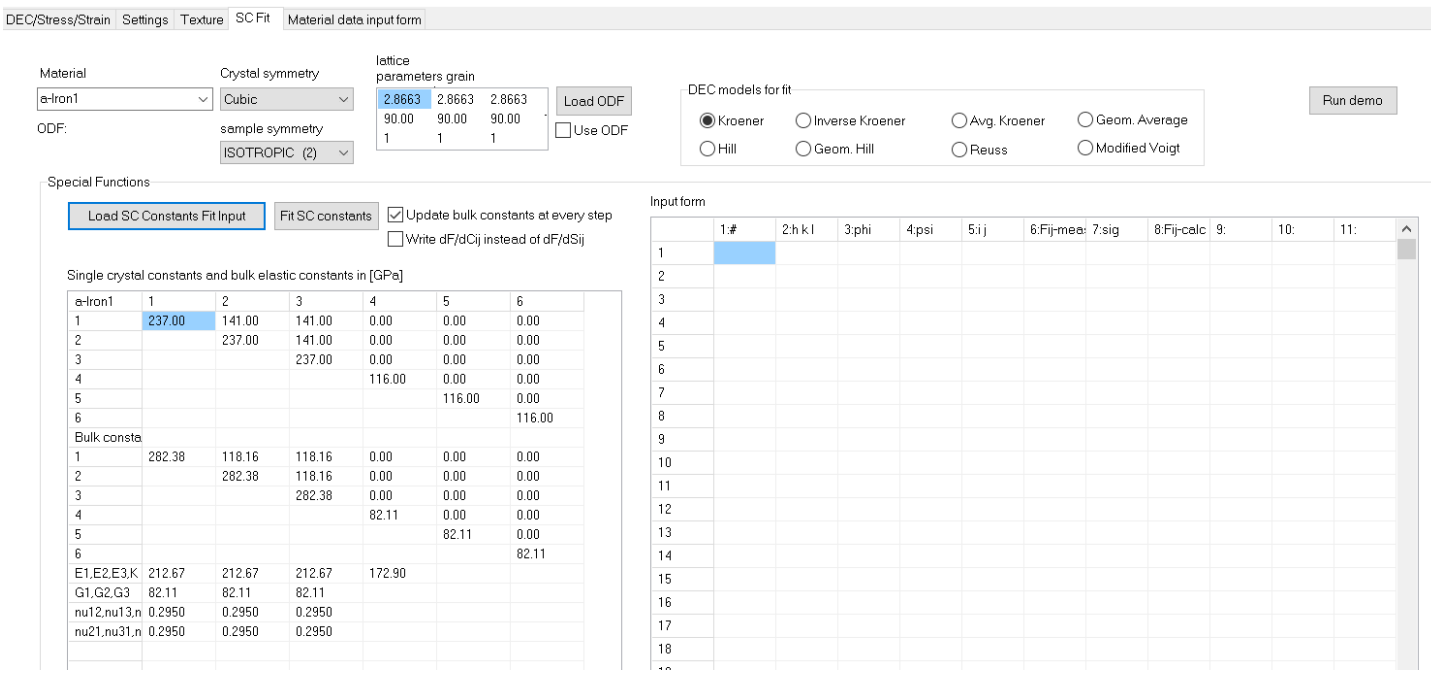
"**Acceptable misorientation**": The primary use of this number is to define the acceptance window for finding grains in EBSD files to match a given orientation. One example would be to find grains that fulfill the diffraction condition for a given (hkl) and ($\varphi\psi$).

"**Alignment of the cartesian frame**": Tensor data such as single crystal elastic constants are based on cartesian bases even of the crystal lattice is not (monoclinic, triclinic). The required cartesian transformation needs a relationship between the crystal base axes and the cartesian axes. The default choice is the alignment of X with the lattice vector **a**, Y is parallel to the reciprocal lattice vector **b***, and Z= X $\times$ Y.

"**Show plot window**": Unchecking the box will minimize the plot window.

6. SC Fit

This tab offers a simple interface for input of diffraction elastic constants and the subsequent fit of single crystal elastic constants in which, through the selected DEC model, DEC are calculated from the single crystal values (left table). The single crystal values are refined in a least squares iteration until the weighted sum of squared differences $\sum_{k=1}^{NPTS} \sigma_k^{-2} \left(F_{ij}^{meas} - F_{ij}^{calc}\right)^2$ converge at a minimum. The built-in demo mode demonstrates how quickly results can be achieved. Input data are provided through the table on the right, with hkl, phi, psi having the usual meaning. Column 5 contains the direction information (tensor indices) in which the stress was applied, and column 6 and 7 contain the measured data and the uncertainties. It must be emphasized again that the F_{ij} are simply the slopes of $\partial\varepsilon(hkl, \varphi\psi)/\partial\sigma_{applied}$, which are usually obtained by linear regression. The scope of the selected DEC model selected on this page is limited to the SC constants fit.



a-Iron1	1	2	3	4	5	6
1	237.00	141.00	141.00	0.00	0.00	0.00
2		237.00	141.00	0.00	0.00	0.00
3			237.00	0.00	0.00	0.00
4				116.00	0.00	0.00
5					116.00	0.00
6						116.00
Bulk consta						
1	282.38	118.16	118.16	0.00	0.00	0.00
2		282.38	118.16	0.00	0.00	0.00
3			282.38	0.00	0.00	0.00
4				82.11	0.00	0.00
5					82.11	0.00
6						82.11
E1,E2,E3,K	212.67	212.67	212.67	172.90		
G1,G2,G3	82.11	82.11	82.11			
nu12,nu13,n	0.2950	0.2950	0.2950			
nu21,nu31,n	0.2950	0.2950	0.2950			

"**Update bulk constants at every step"** : This is the only option that has not been previously discussed. It recalculates the bulk constants after every iteration, and it affects only calculations using DEC models that contain grain-matrix interaction (Kröner type models, Geometric Average). Disabling the option would make sense if the input data were measured on a multi-phase material, or where bulk elastic constants were measured independently.

Output of the fit is written to a file called "SCfit.res". Flow control for the fit is limited in this mode. Only the maximum number of iterations (col 2, row 7) can be edited.

"**Write dF/dCij instead of dF/dSij**" : The partial derivatives of the stress factors (DEC) can be expressed either through the compliances S_{ij} or the stiffnesses C_{ij}. The results are written in 'SCFit.res', and it can be used to quantify the sensitivity of a particular component S_{ij} or C_{ij}. To the measured (hkl). This sensitivity is less important for cubic materials due to the limited number of (hkl) available but it can be helpful for deciding which (hkl) to measure if the goal is to obtain single crystal elastic constants.

6.1.Using an input file

A greater level of control over the fit is achieved through an input file where individual single crystal elastic constants can be fixed or freed in the fit. The input data are structured in the following way:

```
Name: Titanium
Titanium        C11     C16     C15     C12     C14     C13     C66     C56     C26     C46     C36     C55     C25     C45     C35     C22     C24     C23     C44     C34     C33
Cryst-Cij:      162.4   0.0     0.0     92      0.0     69.00   35.2    0.0     0.0     0       0       46.7    0       0       0       162.4   0       69.00   46.7    0       180.7
Cij-Refine:     1       0       0       2       0       3       0       0       0       0       0       4       0       0       0       0       0       0       0       0       5
Macro-Cij:      165.1   0.0     0       78.4    0.0     78.4    43.35   0       0       0       0       43.35   0       0       0       165.1   0       78.4    43.35   0       165.1
Lattice-Param:  2.9508 2.9508  4.6855 90 90 120
Macro-Sym:      0
Crystal-Sym:    2
NPTS:           20
MAXITER:        6
UpdateBulk:     1
Data:
1 0 0   0   0   1 1 -2.825  0.218
1 0 0   0   90  1 1 9.492   0.218
0 0 2   0   0   1 1 -2.106  0.204
0 0 2   0   90  1 1 7.869   0.204
1 0 1   0   0   1 1 -2.881  0.100
1 0 1   0   90  1 1 9.047   0.100
1 0 2   0   0   1 1 -2.535  0.213
1 0 2   0   90  1 1 8.615   0.213
1 1 0   0   0   1 1 -3.124  0.167
1 1 0   0   90  1 1 9.006   0.167
1 0 3   0   0   1 1 -2.523  0.141
1 0 3   0   90  1 1 7.987   0.141
1 1 2   0   0   1 1 -2.981  0.123
1 1 2   0   90  1 1 8.725   0.123
2 0 1   0   0   1 1 -2.804  0.143
2 0 1   0   90  1 1 8.840   0.143
0 0 4   0   0   1 1 -2.009  0.289
0 0 4   0   90  1 1 7.368   0.289
2 0 2   0   0   1 1 -2.659  0.224
2 0 2   0   90  1 1 9.272   0.224
```

Specific data are parsed through keywords (not case sensitive). Keywords are:

Name — identifies the name of the material

Cryst-Cij — Initial starting values for the single crystal elastic constants in GPa. Values must be given in the order specified in line 2. The C_{ij} will be made symmetric according to the crystal symmetry, and in decreasing priority. For example, for the C_{ij} in cubic symmetry one has $C_{11}=C_{22}=C_{33}$, $C_{12}=C_{13}=C_{23}$, and $C_{66}=C_{55}=C_{44}$. After reading the C_{ij} the value C_{22} and C_{33} are set equal to C_{11}, C_{13} and C_{23} are set equal to C_{12} and C_{55} and C_{44} are set equal to C_{66}. The order used here goes back to the 4th rank tensor notation.

Cij-Refine — Specific values Cij are freed by giving it an integer number>0. The only distinction made is zero or non-zero.

Macro-Cij — Bulk elastic constants tensor in GPa. Sample symmetry is applied in the same way as described for the single crystal values. Therefore, strictly speaking, only C_{11} and C_{22} need to be given for isotropic symmetry.

Lattice-Param — Given in the order a b c alpha beta gamma

Macro-Sym — Sample symmetry. The following codes are accepted: 0=isotropic, 1=hexagonal, 2=orthorhombic, 3=monoclinic, 4=triclinic.

Crystal-Sym — Symmetry of the unit cell. The following codes are accepted:
0=Isotropic, (2 independent non-zero components),
1=cubic, (3 independent non-zero components),
2=hexagonal, (5 independent non-zero components),
3= Tetragonal1 (6 independent non-zero components),
4=Tetragonal2 (7 independent non-zero components),
5=Trigonal1 (5 independent non-zero components)
6=Trigonal2 (6 independent non-zero components)
7=Orthorhombic (9 independent non-zero components)
8=Monoclinic1 (Y||b, diad ||b , 13 independent non-zero components)
9=Monoclinic2 (Z||c, diad ||c , 13 independent non-zero components)
10=Triclinic (21 non-zero components)

NPTS — Number of data lines to read in the format:
h k l phi psi i j Fij sigma

	The indices 'i j' indicate the direction of the applied stress, and therefore the indices for the F_{ij}.
Maxiter	Maximum number of iteration loops.
UpdateBulk	A non-zero value here enables the update of the bulk constants tensor at every iteration. It affects only fits where models with grain-matrix interaction were selected (Kröner, inverse Kröner, average Kröner, geometric average)
Data	There must be NPTS lines with data following this keyword.

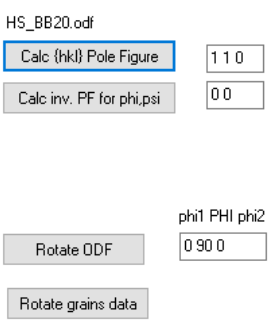
6.2. Output file

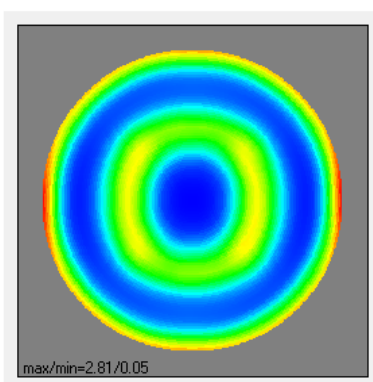
If an input file was provided the output will be written using the input file name with the extension ".res" in append mode. The output will contain fit results with uncertainties (1×σ), Chi-squared ($\chi^2 = \sum_{k=1}^{NPTS} \sigma_k^{-2} \left(F_{ij}^{(k),meas} - F_{ij}^{(k),calc}\right)^2$, reduced chi-squared ($\chi^2_{reduced} = \chi^2/(NPTS - M)$) where M is the number of fit parameters, and σ_k is the uncertainty of $F_{ij}^{(k),meas}$. Further data given are the bulk elastic constants and directional moduli E_1, E_2, E_3, K, nu_{12}, nu_{13}, nu_{23}, nu_{21}, nu_{31}, nu_{32}, G_1, G_2, and G_3. Any directionality would arise from preferred orientation, or if the matrix constants are explicitly anisotropic. The subscripts indicate the direction. The moduli are obtained from the bulk compliance tensor, for example, $E_i = 1/S_{iii}$ and $\upsilon_{ij} = \varepsilon_{jj}/\varepsilon_{ii}$ for stress applied in ii-direction.

Also given are the sensitivities $\partial F_{ij}(hkl, \varphi\psi)/\partial S_{kl}$ which state how sensitive a particular diffraction elastic constant $F_{ij}(hkl, \varphi\psi)$ is to a change in the single crystal compliance S_{kl}. In short, low sensitivities suggest to fix, rather than free, particular parameters in the fit. In practice, similar information can be obtained from the uncertainties of the fitted parameters, and from the correlation coefficients listed in the output file.

7. Texture

This sub-section contains a list of functions of some practical value to texture base calculations.





If an ODF for material 1 is loaded then the functions listed here can be used for a fast display of pole figures and inverse pole figures. The functions "Rotate ODF" and "Rotate grains data" are used if the coordinate system of the diffraction measurement (φ,ψ) is different from the coordinate system in which the ODF or EBSD data were measured. For example, for sheet materials EBSD data are often obtained on the cross-section surface (RD/TD-ND plane) while diffraction measurements are usually done on the sheet surface (RD-TD plane).

Clicking the "Rotate" buttons will perform the rotations using the Bunge angles phi1, PHI, phi2. Once the rotation is performed (see lower right corner of the program window) the user can click the "Calc {hkl} pole figure" again to confirm the results. Plotting convention is RD=East, TD=North.

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